

set_os

REFERENCE

API consumer

initiate an over-the-air firmware update on the reader

Overview

`set_os` initiates an over-the-air (OTA) firmware update — it points the reader at a firmware `.DAT` image at a URL and triggers the download-and-flash cycle. It is a state-changing command on the device-management (MGMT) endpoint, and it is both **asynchronous** and **session-severing**: the device accepts the payload, downloads and flashes the firmware, and **reboots** — dropping off the network for the duration. The authoritative confirmation of the outcome is an out-of-band `get_version` read-back.

`set_os` **reflashes and reboots the reader**. The reply `code 1` is an accept (returned for success and failure alike), and the reader rejects every other command with `code 4` while the update runs. See the danger callout in [Notes and Caveats](#); confirm the result with `get_version`.

Details

Property	Value
Interaction Pattern	Asynchronous Request → Accept → Reboot → Verify (completion confirmed out-of-band)
State Change	STATE-CHANGING (downloads and flashes firmware; reboots the reader)
Reversibility	IRREVERSIBLE (no single-step undo; re-flash a prior build with another <code>set_os</code> to roll back)
Required Request Fields	<code>command</code> , <code>requestId</code> , <code>OSUpdateDetails</code> (with <code>url</code> and <code>authenticationType</code>)
Applies To	Endpoint: Management (MGMT). Reader: RFD40 series, RFD90 series
API Version	<code>v1.1</code> (API Version Reporting)
Related	<code>get_version</code> , <code>install_certificate</code> , <code>reboot</code> , <code>get_status</code>

When to Use

Use `set_os` to **update the reader's firmware** — during commissioning, a fleet migration, or a security/feature rollout.

- **Host the image** — place the firmware `.DAT` at a URL the reader can reach.
- **Choose authentication** — `NONE` for an open/internal server, or `CERTIFICATE` (with a CA reference or inline PEM) plus a `verificationType` for a secured HTTPS download.
- **Check preconditions** — ensure adequate battery and flash first (low battery → code `14`, insufficient flash → code `8`).
- **Verify afterward** — always confirm the new build with `get version`.

Topic Mapping

Message	Direction	Topic	Note
Request	App → Reader	<code><tenantId>/MGMT/clients/cmnd/<serial></code>	On the management endpoint.
Response	Reader → App	<code><tenantId>/MGMT/clients/resp/<serial></code>	The <code>code 1</code> accept, and the <code>code 4</code> lock returned to <i>other</i> commands during the install.
Event	Reader → App	<code><tenantId>/MGMT/clients/event/<serial></code>	The <code>FIRMWARE_UPDATE</code> notifications (start, progress, outcome). See <code>alerts</code> .

Topics follow the pattern `<tenantId>/<topic>/<serial>` — the tenant id assigned to you at provisioning (a slot you fill in, never a literal), the base topic you configure on the endpoint, and the reader's serial number, which the reader appends itself in its model-prefixed form (for example `RFD40-212735201D0053`). `App → Reader` means your app publishes and the reader subscribes; `Reader → App` means the reader publishes and your app subscribes.

See [Topics and Channels](#).

Key Fields

- `OSUpdateDetails.url` (request) — the firmware `.DAT` download URL (a bad/unreachable URL → code `9`).
- `OSUpdateDetails.authenticationType` (request) — `NONE` or `CERTIFICATE`; for `CERTIFICATE`, supply `caCertificateFile` (installed reference) or `caCertificateFileContent` (inline PEM).

- `OSUpdateDetails.verificationType` (request) — TLS verification of the download server; we recommend setting it for HTTPS downloads (see Request Semantics).
- `response.code` (response) — 1 means *accepted*, not *updated*; confirm with `get version`. See [Response Codes](#).

Request Payload

The envelope is `command`, `requestId`, and the `OSUpdateDetails` payload (`url` + `authenticationType`). Pick an **update source** below; the Example and Schema update together. Use the **Schema** tab for the full field reference and the **Example** tab for the concrete payload.

Update source

No authentication HTTP

Request payload

Example: authenticationType NONE · HTTP

```
1 {
2   "command" : "set_os" ,
3   "requestId" : "abc123" ,
4   "OSUpdateDetails" : {
5     "url" : "http://192.168.29.39:8000/Build-3.10.27/Firmware/PAAFKS00-012-R
6     "authenticationType" : "NONE" ,
7     "verificationType" : "VERIFY_NONE"
8   }
9 }
```

Request payload

* required • — not applicable • Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE
1	command *	string	—	set_os
2	requestId *	string	—	abc123
3	OSUpdateDetails *	object	—	—

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE
4	u r l [*]	string	uri	https://192.168.29.39:8000/Build-3.10.27/Firmware
5	a u t h e n t i c a t i o n T y p e [*]	enum	NONE CERTIFICATE	CERTIFICATE
6	ve ri fi ca ti on Ty pe	enum	VERIFY_NONE VERIFY_HOST_AND_PEER	VERIFY_NONE

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE
7	certificate	string	—	<pre> -----BEGIN CERTIFICATE----- \nMIIDPTCCAiWgAwIBAgIUHtj1WDceNR9 1Ne7sFmHkwDQYJKoZIhvcNAQEL\nBQAwI A1UEAwwGVGVzdENBMQ4wDAYDVQQKDAVaZ MAkGA1UEBhMC\nVVMwHhcNMjYwNTEwMTM cNMzYwNTA4MTMxNDQ3WjAuMQ8wDQYDVQQ ZXN0Q0ExDjAMBGNVBAoMBVplYnJhMQswC EwJVUzCCASIwDQYJKoZIhvcN\nAQEBBQA CCAQoCggEBANR8E/mxoQ5uTPW/muWnGoW lIiJiCM/n92\ndMLIQ+TVgb9mcOz80Vnpg ZaIG4CtnBufyIJvNKid+mgrrrFh18HtMnE oPC4M469Rm+0oEN9GkUHgtgF2N4Qt9bYo yPlFuUOC+uaYRCCn3Hm8ko\nXE/a7XWdX Mzzf7MoNq14AfX431RKgvpMgAqZTnDWtm q0t64tP7\naKIAJV2avJd87dF8x+K80kL uNFpXN7KoC7BBwdTzbG8VWYNUZeG8eCqB Y5Ik7POMR0AM/TxRBLAXyoZHSPdEpEoaQ Hv0CAwEAAaNTMFEwHQYD\nVR0OBBYEFJH FYnVcf2++X9N8UlkahMB8GA1UdIwQYMBa XPeyFY\nnnVcf2++X9N8UlkahMA8GA1UdE MAMBAf8wDQYJKoZIhvcNAQELBQADggEB\ MhuQ9sQLFE9wiGw4BIILxnX30FbphWM8/ sH/NOWpjulG+ZAxe8n\n+QUhaU9hruBEa vZ9NtUumcAmAh3s29KD4NpjwglqUYfKQp MXuW\nnzCmqlAqGVzK3ve/Fuurgdbk3kAp w629dmmRhoP+2sdCYMAuky348/4G1\nnY ygC4/WCfWg/7XaeQ+6kqKtY6ijSJFFCNw 3XAqP+kZ0e2taL5a\nnrv5UUPRMGCS20J 8uPOICNWUER3DbqAJ4jXKhSHdsG6fNDyJ 0j\nVqh5mX6k2PZe0eylcukyu/s=\n--- CERTIFICATE----- </pre>
8	certificate	string	—	/apps/ca_cert.pem

Request Semantics

Modes. `authenticationType` selects how the reader authenticates to the firmware source: `NONE` for an unauthenticated download (an open or internal server), or `CERTIFICATE` to authenticate the HTTPS download against a CA — supplied either as a **pre-installed** reference (`caCertificateFile` , installed via `install_certificate`) or **inline** (`caCertificateFileContent`), and paired with a `verificationType` for TLS server verification. Both `CERTIFICATE` forms are in the [Request Payload](#) selector: **Pre-installed CA certificate HTTPS** and **Inline CA certificate HTTPS**.

- **Authentication & transport security.** For `authenticationType: CERTIFICATE` , supply the CA either as an installed reference (`caCertificateFile` , pre-installed via `install_certificate`) or inline (`caCertificateFileContent`); if `caCertificateFileContent` is empty, the reader uses an already installed filestore certificate. `authenticationType: NONE` means no authentication is needed to access the URL. `verificationType` sets how far the reader validates the download server's TLS certificate: `VERIFY_HOST_AND_PEER` checks the certificate chain and that the certificate matches the host; `VERIFY_NONE` does no verification and must not be used in production.
- `OSUpdateDetails` **is required**. The contract requires it, so a request without it is invalid rather than merely unable to update. A bad or unreachable `url` returns code `9` .

Response Payload

The immediate reply is an **accept** — `command` , `requestId` , `apiVersion` , and `response` with `code 1` . It confirms receipt, not a completed update.

Response payload

Example: accept

```
1  {
2      "command" : "set_os" ,
3      "requestId" : "abc1234" ,
4      "apiVersion" : "V1.1" ,
5      "response" : {
6          "code" : 1 ,
7          "description" : "Command payload is accepted"
8      }
9  }
```

Response payload

* required • — not applicable • Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
1	command *	string	—	set_os	Echoes the executed command.
2	requestId *	string	—	abc123	Echoes the request's requestId — pairs reply to request.
3	apiVersion *	enum	V1.0 V1.1 V1.2 V1.21 V1.22	v1.1	API version reported by this command. Emitted per-handler, not device-wide.
4	response	object	—	—	Accept/status envelope. Not in the contract's required set for this reply, although every published example carries it. This is an accept, not a terminal result (see the lifecycle below).
5	code *	integer	0–101	0	1 = the payload was accepted (the accept code for this command — not 0). See Response Codes below.
6	description *	string	—	Success	Human-readable status.

Lifecycle

`set_os` is an accept-then-verify cycle, not a single request/response.

Acknowledgment

The device replies `response.code 1` ("Command payload is accepted") on the response topic. This confirms receipt only, and is **identical for success and failure**.

Progress

A `FIRMWARE_UPDATE` notification appears on the event topic as the download/flash begins. The block below is the example the schema declares for it, not an observed one:

Event — FIRMWARE_UPDATE start (schema example)

```
1  {
2    "type": "NOTIFICATION",
3    "timestamp": "2026-06-18T11:30:50.387Z",
4    "state": "ONESHOT",
5    "id": "FIRMWARE_UPDATE",
6    "priority": "HIGH",
7    "alertDetails": {
8      "fwUpdateStatus": {
9        "updateStatus": "Starting FW update"
10     }
11  }
12 }
```

Notifications continue on the same topic while the install runs. Each of the five recorded during one update observed on an RFD40 carried `alertDetails.fwUpdateStatus.overallProgress` — percent of the update completed — beside a human-readable `updateStatus`. `overallProgress` is optional and the start example above carries none, so treat its absence as unknown rather than as zero. Three of those five notifications are shown here and under Confirmation below, across the 15.265 s in which `overallProgress` ran from 87% to 100%; the earlier part of that update was not recorded.

Event — FIRMWARE_UPDATE in progress, 87% (live)

```
1  {
2    "type": "NOTIFICATION",
3    "timestamp": "2026-07-07T11:49:50.628Z",
4    "state": "ONESHOT",
5    "id": "FIRMWARE_UPDATE",
6    "priority": "HIGH",
7    "alertDetails": {
8      "fwUpdateStatus": {
9        "updateStatus": "In progress",
10       "overallProgress": 87
11     }
12  }
13 }
```

The identical `In progress` value arrived twice — at 87%, then at 95% 10.014 s later — so only the first is shown. The notification after it carried a status string naming a component file inside the firmware package:

Event — FIRMWARE_UPDATE in progress, 98% (live)

```
1  {
2    "type": "NOTIFICATION",
3    "timestamp": "2026-07-07T11:50:04.424Z",
4    "state": "ONESHOT",
5    "id": "FIRMWARE_UPDATE",
6    "priority": "HIGH",
7    "alertDetails": {
8      "fwUpdateStatus": {
9        "updateStatus": "Updating file PAAFKS00-013-R04L0.DAL",
10       "overallProgress": 98
11     }
12   }
13 }
```

- `updateStatus` is display text, not a machine token. It carries whatever the firmware shows an operator, file names included, and four of the five recorded notifications carried a value that is not one of the five `fwUpdateStatus.updateStatus` declares — only the terminal `FW update success` matched. The declared lowercase values `updating`, `skipping` and `failed` appeared in none of them: the reader sent sentences such as `Updating file ...` and `Skipping file ...` instead, so the obvious test `updateStatus === 'updating'` matches nothing this reader sends. Render the string as opaque text and drive logic from `overallProgress` and from the terminal literal under Confirmation below. The schema remains normative; this is an observed device deviation, cataloged in [Known device deviations from the schema](#).
- **A status string can arrive incomplete.** The notification at 99% reported `Skipping file ...`, naming a path inside the firmware package, and the value ended mid-name, its final character recorded as an encoding-error artifact rather than as text. Whether the reader truncated the string or the recording chain corrupted it cannot be told from a single sample, and no maximum length can be inferred from one either: do not assume a length limit, and do not substring-match. `overallProgress` in that same notification was a well-formed `99`, which is the practical case for branching on the number rather than on the text.
- **The stream's timing is not a contract.** Across the recorded window the gap between notifications ran from 10.014 s down to 0.284 s while `overallProgress` advanced by 8, then 3, then 1, then 1 percentage points — neither a fixed interval nor a fixed increment, so do not derive a silence timeout from it. Emission is gated by `eventConfiguration.firmwareUpdate` in [config_events](#) and can be switched off altogether. The stream therefore gives you no deadline, so bound the wait on your own clock: set a wall-clock budget for the whole update from your own fleet's measurements, and on expiry poll [get_version](#) until the reader answers. Treat silence as unknown, never as failure.

Session Impact

While the update runs, the reader rejects **every other command** on the response topic with `code 4` ("Firmware update in progress"), echoing the blocked command's name:

Response — another command blocked during the update (code 4)

```
1  {
2    "command" : "get_endpoint_config" ,
3    "requestId" : "abc123" ,
4    "apiVersion" : "V1.1" ,
5    "response" : {
6      "code" : 4 ,
7      "description" : "Firmware update in progress"
8    }
9  }
```

The reader reflashes and reboots, and is briefly off the network.

Confirmation

The reader also reports the outcome on the event topic. The terminal notification carries `updateStatus: FW update success` with `overallProgress: 100`:

Event — FIRMWARE_UPDATE success, 100% (live)

```
1  {
2    "type" : "NOTIFICATION" ,
3    "timestamp" : "2026-07-07T11:50:05.893Z" ,
4    "state" : "ONESHOT" ,
5    "id" : "FIRMWARE_UPDATE" ,
6    "priority" : "HIGH" ,
7    "alertDetails" : {
8      "fwUpdateStatus" : {
9        "updateStatus" : "FW update success" ,
10       "overallProgress" : 100
11      }
12    }
13  }
```

That literal is the one `updateStatus` value worth an equality test — it is a declared enum member and the observed notification matched it exactly — with `overallProgress: 100` as corroboration rather than as the test, because a single observation cannot establish that 100 always means success. Every notification recorded, the terminal one included, carried `state: ONESHOT`, so the terminal signal sits in `fwUpdateStatus` and not in the envelope: **do not wait for a `CLEAR`**.

Gate on the read-back, not on the notification. Run `get_version` and compare `readerVersion.firmwareVersion` against the target build (see the version note below). The notification tells you when to stop waiting; the read-back tells you whether the firmware changed. A failed update leaves the prior version in place; terminal failure is reported as `fwUpdateStatus.updateStatus: failed` in the `FIRMWARE_UPDATE` event (which carries no response code), not on the `set_os` accept. `failed` is the value the schema declares for that case; no failure arose in the traffic behind the examples above, and because two of the three bare-stem values the schema declares arrive as sentences on this firmware, a real failure may well arrive as a sentence too — so detect failure by the read-back, not by matching `failed`.

Response Codes

`set_os` returns `response.code`. The codes applicable to this command are below; for the full table see the [Response Codes](#) and [Correlation and Request/Response Pairing](#).

Code	Description
1	Command payload is accepted (<code>IOT_STATUS_CMD_PAYLOAD_ACCEPTED</code>) — the accept code for this command . The update proceeds asynchronously; verify with <code>get_version</code> .
0	Success — terminal success; in practice <code>set_os</code> acks with 1, and completion is confirmed via the <code>get_version</code> read-back.
4	Firmware update in progress (<code>IOT_STATUS_FW_UPDATE_IN_PROGRESS</code>) — returned to other commands while an update is running (the device-wide lock). Pause orchestration until the device returns.
8	Insufficient flash size (<code>IOT_ERROR_LOW_FLASH_SIZE</code>) — not enough free flash to download the firmware. Free space or use a smaller package.
9	File not found (<code>IOT_STATUS_FILE_NOT_FOUND</code>) — the firmware <code>url</code> is wrong or unreachable.
13	Firmware update Failed (<code>IOT_ERROR_FW_UPDATE_FAIL</code>) — the defined failure code. In practice <code>set_os</code> acks only with 1; a failed update is reported as <code>fwUpdateStatus.updateStatus: failed</code> in the <code>FIRMWARE_UPDATE</code> event (which carries no response code), not on the <code>set_os</code> response — but do not equality-test that literal; see Confirmation .
14	Battery is low, Cannot update firmware (<code>IOT_ERROR_FW_UPDATE_FAIL_LOW_BATTERY</code>) — charge or connect external power, then retry.

Code	Description
101	Default error — generic catch-all.

Gate success strictly on the `get_version` read-back, not on the `set_os` response code — the reply `code 1` acknowledges only that the payload was accepted, and is returned for both success and failure.

Examples

Publish `set_os`; the device acks with the `code 1` envelope under [Response Payload](#), then runs the [Lifecycle](#). The `CERTIFICATE` variants are in the [Request Payload](#) selector; the `FIRMWARE_UPDATE` notifications — start, in-flight progress, and terminal success — and the `code 4` lock are shown under the Lifecycle above.

Notes and Caveats

⚠ REFLASHES AND REBOOTS THE READER — PLAN FOR UNREACHABILITY

The device goes dark and locks out other commands. While the update runs, the reader rejects **every** other command with `code 4` ("Firmware update in progress") and is then unreachable across the reboot. Do not retry-storm — pause orchestration until the device re-attaches. The reply `code 1` is returned for **both** success and failure, and there is no terminal result on the response topic: confirm the outcome with `get_version` and compare the firmware version.

- **Preconditions.** Ensure the device is ready before issuing: low battery → `code 14`; insufficient flash → `code 8`.
- **Confirm by full version string.** `readerVersion.firmwareVersion` is a free-form string. Compare the **whole** value against the target build — the build-variant prefix can change (e.g. `PAAFKS00-013-R02` → `SAAFKS00-013-R02`), so match the full string rather than a substring. See `get_version`.

Sequence a firmware rollout as: pre-check with `get_status` (power/battery) → `set_os` → wait out the reboot (expect `code 4` on other commands) → confirm with `get_version`. For the concept-level view, see [Update Firmware and Restart the Reader](#).

Related: [Update Firmware and Restart the Reader](#) · [get_version](#) · [install_certificate](#) · [reboot](#) · [Response Codes](#)